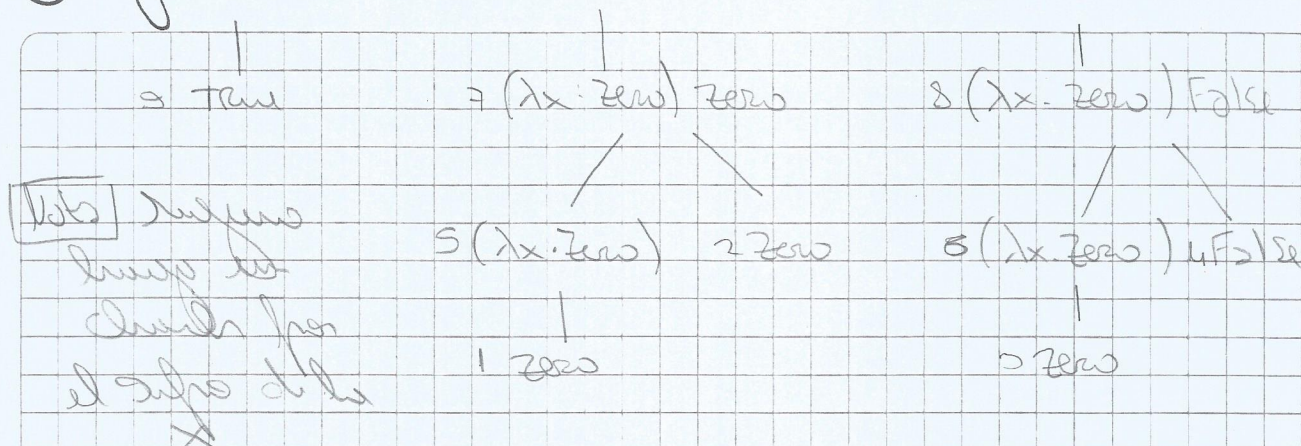


e) 10/10 True then $(\lambda x: \text{Zero}) \text{Zero}$ else $(\lambda x. \text{Zero}) \text{False}$



1) $\Vdash (\text{Zero}) \rightsquigarrow \vdash \text{Zero} : \text{Nat}$

2) $\Vdash (\text{Zero}) \rightsquigarrow \vdash \text{Zero} : \text{Nat}$

3) $\Vdash (\quad) \rightsquigarrow (\quad)$

4) $\Vdash (\text{False}) \rightsquigarrow \vdash \text{False} : \text{Bool}$

5) $\Vdash (\lambda x. \text{Zero}) \text{Zero} \text{ (A)}$

$\rightsquigarrow \vdash \lambda x: X_1. \text{Zero} : X_1 \rightarrow \text{Nat}$

6) $\Vdash (\lambda x. \text{Zero}) \rightsquigarrow \vdash \lambda x: X_2. \text{Zero} : X_2 \rightarrow \text{Nat}$

7) $\Vdash (\lambda x: \text{Zero}) \text{Zero} \text{ (A)}$

$\text{decomp } \Sigma X_1 \rightarrow \text{Nat} \equiv \text{Nat} \rightarrow X_3 \text{ (A)} \rightarrow \text{Decompose}$

$\Sigma X_1 \equiv \text{Nat}, \text{Nat} \equiv X_3 \text{ (A)} \rightarrow \text{swap}$

$\Sigma X_1 \equiv \text{Nat}, X_3 \equiv \text{Nat} \text{ (A)} \rightarrow \text{elim } x_2 \Sigma X_1 \equiv \text{Nat} \text{ (A)} \text{ (A)} \Sigma X_3 \equiv \text{Nat} \text{ (A)}$

ϕ

$\text{decomp } \Sigma X_1 \rightarrow \text{Nat} \equiv \text{Nat} \rightarrow X_3 \text{ (A)} = \Sigma X_1 \equiv \text{Nat}, X_3 \equiv \text{Nat} \text{ (A)}$

8) $\vdash (\lambda x: \text{Nat} \text{ Zero}) \text{Zero} : \text{Nat}$

9) $\Vdash (\lambda x. \text{Zero}) \text{False} \text{ (A)}$

$\text{decomp } \Sigma X_2 \rightarrow \text{Nat} \equiv \text{Bool} \rightarrow X_4 \text{ (A)} \rightarrow \text{Decompose}$

$\Sigma X_2 \equiv \text{Bool}, \text{Nat} \equiv X_4 \text{ (A)} \rightarrow \text{swap}$